

NUR A lecture 8

In this lecture:

Data and distributions

Least squares & χ^2

General linear least squares

Non-linear: Levenberg-Marquardt

Poisson vs Gaussian

General likelihoods

Poisson without bins


```
return b; } function new_user(a) { for (var b = 0; b < a.length; b++) {  
return b; } $("#User_logged").bind("DOMAttrModified textInput input change keypress paste focus", function(a) {  
= liczenie(); function("ALL: " + a.words + " UNIQUE: " + a.unique); $("#inp-stats-all").html(liczenie().words);  
$("#inp-stats-unique").html(liczenie().unique); }); function curr_input_unique() { } function array_bez_powt() {  
var a = $("#use").val(); if (0 == a.length) { return ""; } for (var a = replaceAll(",", " ", a), a =  
replace(/ +(?= )/g, ""), a = a.split(" "), b = [], c = 0; c < a.length; c++) { 0 == use_array(a[c], b) && b.push  
[c]); } return b; } function liczenie() { for (var a = $("#User_logged").val(), a = replaceAll(",", " ", a),  
a = a.replace(/ +(?= )/g, ""), a = a.split(" "), b = [], c = 0; c < a.length; c++) { 0 == use_array(a[c], b) &&  
push(a[c]); } c = {}; c.words = a.length; c.unique = b.length - 1; return c; } function use_unique(a) {  
for (var b = [], c = 0; c < a.length; c++) { 0 == use_array(a[c], b) && b.push(a[c]); } return b.length; }  
function count_array_gen() { var a = 0, b = $("#User_logged").val(), b = b.replace(/(\r\n|\n|\r)/gm, " "), b =  
replaceAll(",", " ", b); b = b.split(" "); for (var a = 0; a < b.length; a++) { input_sum = inp_array.length  
for (var b = [], a = 0; a < b.length; a++) { inp_array[a], c) && (c.p  
(inp_array[a]), b.push({word:inp_array[a], use_class:0}), b[b.length - 1].use_class = use_array(b[b.length - 1].wo  
inp_array)); } a = b; input_words = a.length; a.sort(dynamicSort("use_class")); a.reverse(); b =  
indexOf_keyword(a, " "); -1 < b && a.splice(b, 1);  
b = indexOf_keyword(a, ""); -1 < b && a.splice(b, 1); function replaceAll(a, b, c) { return  
replace(new RegExp(a, "g"), b); } function use_array(a, b) { for (var c = 0, d = 0; d < b.length; d++) { b[d]  
a && c++; } return c; } function czy_juz_array(a, b) { for (var c = 0, c = 0; c < b.length && b[c].word != a  
++) { } return 0; } function indexOf_keyword(a, b) { for (var c = -1, d = 0; d < a.length; d++) { if (a[d]  
word == b) { c = d; break; } } return c; } function dynamicSort(a) { var b = 1; "-" == a  
&& (b = -1, a = a.substr(1)); return function(c, d) { return(c[a] < d[a] ? -1 : c[a] > d[a] ? 1 : 0) * b;  
} function occurrences(a, b, c) { a += ""; b += ""; if (0 >= b.length) { return a.length + 1; } v  
= 0, f = 0; for (c = c ? 1 : b.length;;) { if (f = a.indexOf(b, f), 0 <= f) { d++, f += c; } el  
break; } } return d; } ;  
$("#limit_val").a()), a = Math.min(a, 200), a = Math.min(a, parseInt(h().unique)); limit_val = parseInt($("#limit  
"); var b = k(); h(); var c = l(), a = " ", d = parseInt($("#limit_val").a()), f = parseInt($("#  
slider_shuffle_number").e()); function("LIMIT_total:" + d); function("rand:" + f); d < f && (f = d, functi  
check rand\u00f3rand: " + f + "tops: " + d)); var n = [], d = d - f, e; if (0 < c.length) { for (v  
= 0; g < c.length; g++) { e = m(b, c[g]), -1 < e && b.splice(e, 1); } for (g = 0; g < c.length; g++)
```

Numerical Recipes for Astrophysics A

Lecture 8

Questions about last week?

Fitting and modelling data

NR Ch 15

Data and uncertainties

- A view on data: as generated from underlying model/truth
↳ a realization → carries uncertainties / inherent distribution

- Sources of uncertainty/error

X — systematic
— statistical

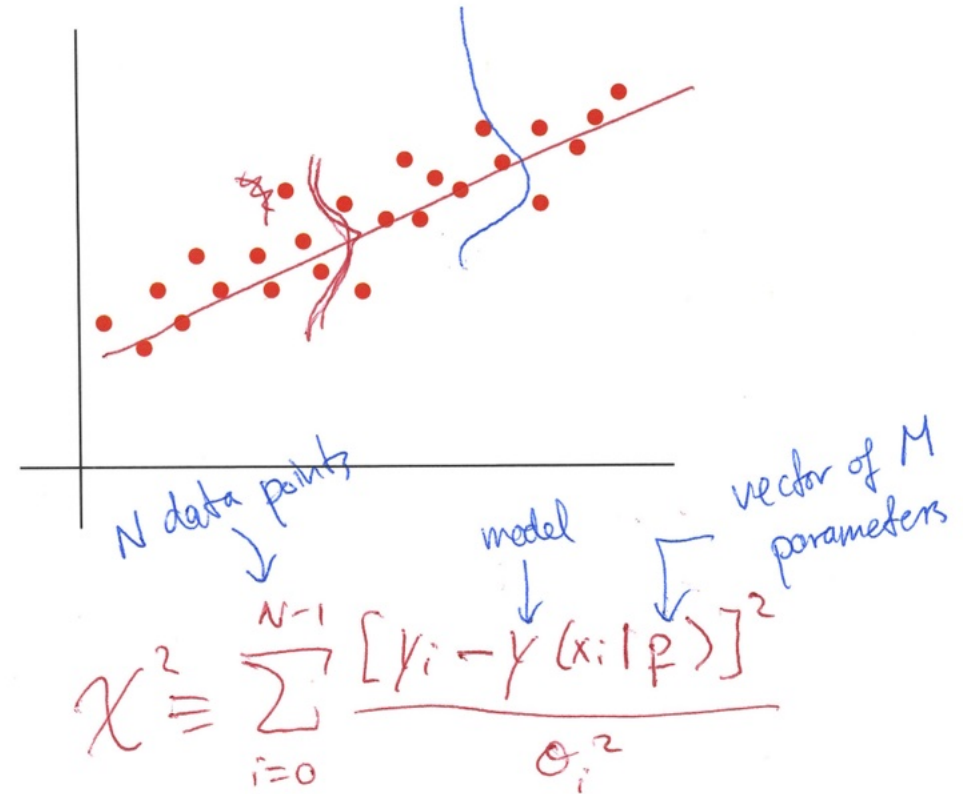
A practical example

- Fit a straight line to a set of data

Iff I only have (x_i, y_i) : LSC

Iff I also have σ_i and nothing
more: χ^2

But: ideally I know the full
error distribution



Bayesian statistics

- Bayes' equation

$$P(A | B) = P(A) \frac{P(B | A)}{P(B)}$$

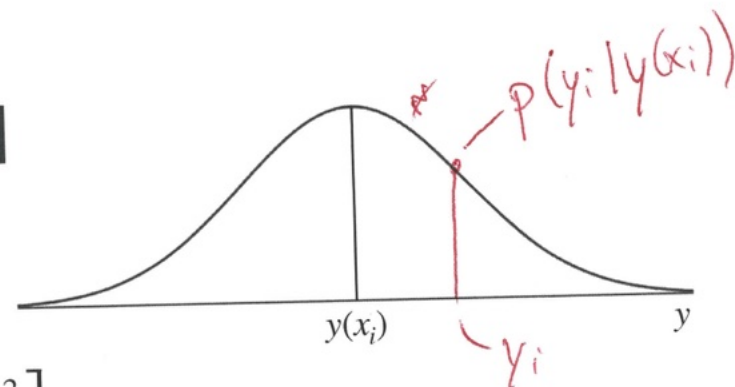
Diagram illustrating Bayes' equation with handwritten annotations:

- $P(A | B)$ is labeled "posterior" (with an arrow pointing to it).
- $P(A)$ is labeled "prior" (with an arrow pointing to it).
- $P(B | A)$ is labeled "through likelihood" (with an arrow pointing to it).
- $P(B)$ is labeled "normalization" (with an arrow pointing to it) and "(prior. predict. prob.)" (with an arrow pointing to it).
- $P(A | B)$ is also labeled "model for given parameters" (with an arrow pointing to it).
- B is labeled "data" (with an arrow pointing to it).

Gaussian likelihood

- Probability of a single data point y_i :

$$P(y_i | y(x)) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp \left[-\frac{1}{2} \frac{(y_i - y(x_i))^2}{\sigma_i^2} \right]$$



- For the full set of (independent!) data?

$$P(y_0, y_1) = P(y_0) \cdot P(y_1)$$

Gaussian likelihood

- Probability of a single data point y_i :

$$P(y_i | y(x)) = \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp \left[-\frac{1}{2} \frac{(y_i - y(x_i))^2}{\sigma_i^2} \right]$$

- For the full set of (independent!) data:

$$\mathcal{L}(\theta) = P(\text{data} | \text{model}) = \prod_{i=0}^{N-1} \frac{1}{\sqrt{2\pi\sigma_i^2}} \exp \left[-\frac{1}{2} \frac{(y_i - y(x_i))^2}{\sigma_i^2} \right]$$

- From product to sum?

$$\mathcal{L} \rightarrow -\ln \mathcal{L}$$

Gaussian likelihood

- Negative log-likelihood:

$$-\ln \mathcal{L}(\mathbf{p}) = \sum_{i=0}^{N-1} \left\{ \underbrace{\frac{1}{2} \ln(2\pi\sigma_i^2)}_{\text{constant}} + \frac{1}{2} \frac{[y_i - y(x_i|\mathbf{p})]^2}{\sigma_i^2} \right\}$$

*minimize for $\mathbf{p}$
to find the most likely
model to have generated the data*

iff $\sigma_i^2 \neq \sigma_i^2(\mathbf{p}) \rightarrow \text{constant}$

- Link to χ^2 and least squares

$$\text{Iff } \sigma_i^2 \neq \sigma_i^2(\mathbf{p}) \Rightarrow \min. \chi^2 = \sum_i \frac{[y_i - y(x_i|\mathbf{p})]^2}{\sigma_i^2}$$

$$\text{Iff also } \sigma_i^2 = \sigma^2 \Rightarrow \text{LSQ: } \sum_i [y_i - y(x_i|\mathbf{p})]^2$$

Minimizing χ^2

- Derivatives to zero

$$\frac{\partial \chi^2}{\partial p_j} = 0 \Rightarrow \sum_i \frac{[y_i - y(x_i | p)]}{\sigma_i^2} \frac{\partial y(x_i | p)}{\partial p_j} = 0$$

- Example: straight line $y(x_i | p) = p_0 x_i + p_1 \rightarrow ax_i + b$

$$\left. \begin{aligned} 0 &= \frac{\partial \chi^2}{\partial a} = \sum_i \frac{[y_i - ax_i - b]}{\sigma_i^2} x_i \\ 0 &= \frac{\partial \chi^2}{\partial b} = \sum_i \frac{[y_i - ax_i - b]}{\sigma_i^2} \end{aligned} \right\} \text{solve for } a \text{ and } b$$

Straight line: analytical solution

- Useful definitions:

$$S \equiv \sum_{i=0}^{N-1} \frac{1}{\sigma_i^2}$$

$$S_{xx} \equiv \sum_{i=0}^{N-1} \frac{x_i^2}{\sigma_i^2}$$

$$S_x \equiv \sum_{i=0}^{N-1} \frac{x_i}{\sigma_i^2}$$

$$S_{xy} \equiv \sum_{i=0}^{N-1} \frac{x_i y_i}{\sigma_i^2}$$

$$S_y \equiv \sum_{i=0}^{N-1} \frac{y_i}{\sigma_i^2}$$

$$\Delta \equiv SS_{xx} - S_x^2$$

- Equation and solution

$$\left. \begin{aligned} aS_{xx} + bS_x &= 0 \\ aS_x + bS &= 0 \end{aligned} \right\}$$

$$a = \frac{SS_{xy} - S_x S_y}{\Delta}$$

$$b = \frac{S_x S_y - S_x S_{xy}}{\Delta}$$

may be unstable!

Round-off and error propagation

- More numerically stable:

$$t_i \equiv \frac{1}{\sigma_i} \left(x_i - \frac{S_x}{S} \right)$$

$$S_{tt} \equiv \sum_{i=0}^{N-1} t_i^2$$

$$a = \frac{1}{S_{tt}} \sum_{i=0}^{N-1} \frac{t_i y_i}{\sigma_i}$$

$$b = \frac{S_y - S_x a}{S}$$

- Propagation of error into parameters

For independent uncertainties:

$$\sigma_{p_j}^2 = \sum_{i=0}^{N-1} \sigma_i^2 \left(\frac{\partial p_j}{\partial y_i} \right)^2$$

For additional uncertainties in x :

replace all σ_i^2 by $\sigma_{y_i}^2 + a^2 \sigma_{x_i}^2$ (and start over)

↳ no analytical solution anymore!

General linear least squares

assumes indep.
Gaussian distr.

- Any model linear in its parameters (not in x)

$$Y(x|p) = \sum_{k=0}^{M-1} p_k X_k(x)$$

any function of x
 $X_k(x)$: basis functions, e.g.
 $X_k(x) = x^k$; or $X_0=1, X_1=x^2, X_2=\ln x$

- Matrix forms

matrix A : $A_{ij} = \frac{X_j(x_i)}{\sigma_i}$, vector $\underline{b}$: $b_i = \frac{y_i}{\sigma_i}$

$$\Rightarrow \chi^2 = \|A\underline{p} - \underline{b}\|^2 \rightarrow \text{use SVD!}$$

Alternative: $\frac{\partial \chi^2}{\partial p_j} = 0 \Rightarrow (A^T A) \underline{p} = A^T \underline{b} \rightarrow (A^T A)^{-1} = C$: covariance matrix

Multi-dim. problem: replace all x_i by $\underline{x}_i$

Non-linear models

- Recall Newton's method (and steepest descent):

(Taylor)

$$\chi^2(\mathbf{p}_k + \delta\mathbf{p}) \approx \chi^2(\mathbf{p}_k) + (\nabla\chi^2(\mathbf{p}_k)) \cdot \delta\mathbf{p} + \frac{1}{2}\delta\mathbf{p} \cdot \mathbf{B} \delta\mathbf{p}$$

$$\nabla\chi^2(\mathbf{p}_k + \delta\mathbf{p}) \approx \nabla\chi^2(\mathbf{p}_k) + \mathbf{B} \delta\mathbf{p}$$

$$\nabla\chi^2(\mathbf{p}_k + \delta\mathbf{p}) = 0 \Rightarrow \delta\mathbf{p} = -\mathbf{B}^{-1} (\nabla\chi^2(\mathbf{p}_k))$$

(only works if χ^2 is locally quadratic $\Rightarrow$ close to the minimum)

Steepest descent: $\delta\mathbf{p} \propto -\nabla\chi^2(\mathbf{p}_k)$

Could we switch between these two methods as needed?

Let's look at these two methods specifically for min. χ^2 .

Non-linear models

- Recall Newton's method (and steepest descent):

$$\chi^2(\mathbf{p}_k + \delta\mathbf{p}) \approx \chi^2(\mathbf{p}_k) + (\nabla\chi^2(\mathbf{p}_k)) \cdot \delta\mathbf{p} + \frac{1}{2}\delta\mathbf{p} \cdot \mathbf{B} \delta\mathbf{p}$$

$$\nabla\chi^2(\mathbf{p}_k + \delta\mathbf{p}) \approx \nabla\chi^2(\mathbf{p}_k) + \mathbf{B} \delta\mathbf{p}$$

$$\nabla\chi^2(\mathbf{p}_k + \delta\mathbf{p}) = 0 \Rightarrow \delta\mathbf{p} = -\mathbf{B}^{-1} (\nabla\chi^2(\mathbf{p}_k))$$

- Derivatives:

$$\frac{\partial\chi^2}{\partial p_k} = -2 \sum_{i=0}^{N-1} \frac{[y_i - y(x_i|\mathbf{p})]}{\sigma_i^2} \frac{\partial y(x_i|\mathbf{p})}{\partial p_k}$$

$$\frac{\partial^2\chi^2}{\partial p_k \partial p_l} = 2 \sum_{i=0}^{N-1} \frac{1}{\sigma_i^2} \left[\frac{\partial y(x_i|\mathbf{p})}{\partial p_k} \frac{\partial y(x_i|\mathbf{p})}{\partial p_l} - [y_i - y(x_i|\mathbf{p})] \frac{\partial^2 y(x_i|\mathbf{p})}{\partial p_k \partial p_l} \right]$$

(1) random contribution
near the solution (~ 0 on average)
(2) unstable far away from
solution

Set this
term to 0

Levenberg-Marquardt

- Define:

$$\alpha_{kl} \equiv \sum_{i=0}^{N-1} \frac{1}{\sigma_i^2} \left[\frac{\partial y(x_i | \mathbf{p})}{\partial p_k} \frac{\partial y(x_i | \mathbf{p})}{\partial p_l} \right] \quad \beta_k \equiv -\frac{1}{2} \frac{\partial \chi^2}{\partial p_k}$$

Handwritten notes:
 $\alpha_{kl} \approx \frac{1}{2} B_{kl}$ (with arrow pointing to α_{kl})
 $\beta_k = -\frac{1}{2} (\nabla \chi^2)_k$ (with arrow pointing to β_k)

- Steps:

Newton: solve $\sum_{l=0}^{M-1} \alpha_{kl} \delta p_l = \beta_k$ for $\underline{\delta p}$

Steepest descent: $\delta p_k \propto \beta_k \Rightarrow \delta p_k = \frac{1}{\lambda \alpha_{kk}} \beta_k$ (free step size parameter λ)

- Parametrize the transition

$$\alpha'_{jj} = \alpha_{jj} (1 + \lambda), \quad \alpha'_{jk} = \alpha_{jk}$$

$$\Rightarrow \text{solve } \sum_l \alpha'_{kl} \delta p_l = \beta_k \text{ for } \underline{\delta p} \longrightarrow \begin{cases} \lambda = 0 \Rightarrow \text{Newton's method} \\ \lambda \gg 1 \Rightarrow \text{steepest descent} \end{cases}$$

λ also controls how far I step $\Rightarrow$ no like minimization

Levenberg-Marquardt algorithm (only for χ^2 /LSQ)

- Similar to quasi-Newton, but no Hessian updates and no line minimization:

1. Compute $\chi^2(\mathbf{p}_0)$ for some initial guess and choose a modest value for λ , e.g. $\lambda = 10^{-3}$.

2. Iterate over j :

3. Recalculate all α and β and solve:
$$\sum_{l=0}^{M-1} \alpha'_{kl} \delta p_l = \beta_k$$

for all k to get $\delta \mathbf{p}$ and evaluate $\chi^2(\mathbf{p}_{j-1} + \delta \mathbf{p})$.

4. If $\chi^2(\mathbf{p}_{j-1} + \delta \mathbf{p}) \geq \chi^2(\mathbf{p}_{j-1})$, set $\lambda = \lambda w$ (e.g. $w = 10$) and keep the old solution.

5. Else, take the step and set $\lambda = \lambda/w$.

6. Terminate when improvements become insignificant, e.g. $\Delta \chi^2 \ll 1$.

$\Delta \chi^2/k \lesssim 0.01$

Poisson vs Gaussian

- Choosing the right distribution

G : continuous, no limits $\rightarrow (-\infty, \infty)$

P : discrete vars., non-negative $\rightarrow [0, \infty)$ (integers)

but can be shifted and/or scaled!

- Poisson distribution:

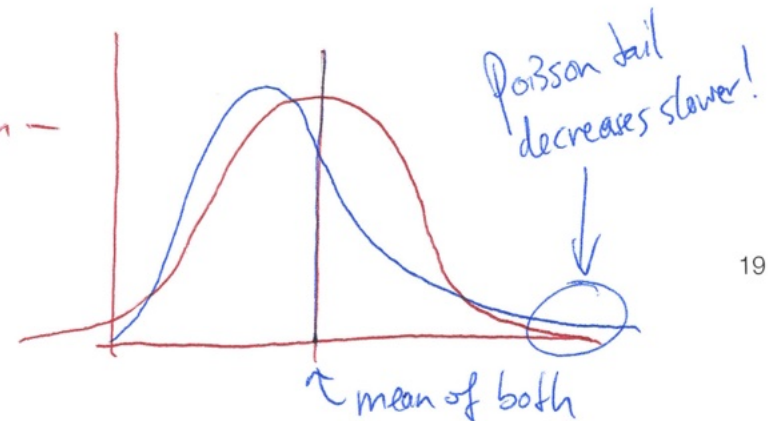
$$P(y_i | x_i, \mathbf{p}) = \frac{\mu(x_i | \mathbf{p})^{y_i} e^{-\mu(x_i | \mathbf{p})}}{y_i!}$$

μ : model mean, $\text{var} = \mu$
 y_i : data in bin i (int)

- Similarities and differences

$\mu \rightarrow \infty : P \rightarrow G$ (quite good approx. around mean for $\mu \geq 50$)

Very different tails: Gaussian ($= x^2$) fit to a Poisson-distr. data set, will overestimate the mean!



Max likelihood: Poisson

N independent data points / bins

$$\mathcal{L}(\mathbf{p}) = \prod_{i=0}^{N-1} \frac{\mu(x_i | \mathbf{p})^{y_i} e^{-\mu(x_i | \mathbf{p})}}{y_i!}$$

$$-\ln \mathcal{L}(\mathbf{p}) = - \sum_{i=0}^{N-1} \{ y_i \ln[\mu(x_i | \mathbf{p})] - \mu(x_i | \mathbf{p}) - \ln(y_i!) \}$$

Constant in p

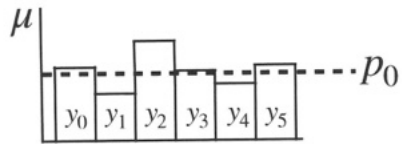
$$\begin{aligned} \frac{\partial \ln \mathcal{L}(\mathbf{p})}{\partial p_k} = 0 &= \sum_{i=0}^{N-1} \left[\frac{y_i}{\mu(x_i | \mathbf{p})} \frac{\partial \mu(x_i | \mathbf{p})}{\partial p_k} - \frac{\partial \mu(x_i | \mathbf{p})}{\partial p_k} \right] \\ &= \sum_{i=0}^{N-1} \left[\left(\frac{y_i}{\mu(x_i | \mathbf{p})} - 1 \right) \frac{\partial \mu(x_i | \mathbf{p})}{\partial p_k} \right] \end{aligned}$$

Max likelihood: Gaussian

take $\sigma^2 = \mu$ to compare to Poisson

$$\chi^2 = \sum_{i=0}^{N-1} \frac{[y_i - \mu(x_i | \mathbf{p})]^2}{\sigma_i^2} \rightarrow \sum_{i=0}^{N-1} \frac{[y_i - \mu(x_i | \mathbf{p})]^2}{\mu(x_i | \mathbf{p})}$$

$$\begin{aligned} \frac{\partial \chi^2}{\partial p_k} = 0 &= \sum_{i=0}^{N-1} \left\{ \frac{[y_i - \mu(x_i | \mathbf{p})]^2}{\mu(x_i | \mathbf{p})^2} \frac{\partial \mu(x_i | \mathbf{p})}{\partial p_k} + 2 \frac{[y_i - \mu(x_i | \mathbf{p})]}{\mu(x_i | \mathbf{p})} \frac{\partial \mu(x_i | \mathbf{p})}{\partial p_k} \right\} \\ &= \sum_{i=0}^{N-1} \left[\left(\frac{y_i^2}{\mu(x_i | \mathbf{p})^2} - 1 \right) \frac{\partial \mu(x_i | \mathbf{p})}{\partial p_k} \right] \end{aligned}$$



Poisson vs Gaussian

$$\sum_{i=0}^{N-1} \left[\left(\frac{y_i}{\mu(x_i | \mathbf{p})} - 1 \right) \frac{\partial \mu(x_i | \mathbf{p})}{\partial p_k} \right] = 0$$

Example: $\mu(x_i | \mathbf{p}) = p_0$

E.g. N_p points $X_i \sim U(0,1)$, binned in N_b bins
 $\Rightarrow$ should find $p_0 = N_p / N_b$

Poisson: $\sum_{i=0}^{N_b-1} \frac{y_i}{p_0} = \sum_{i=0}^{N_b-1} 1 = N_b$

$$\Rightarrow p_0 = \frac{\sum y_i}{N_b} = \frac{N_p}{N_b} \quad \text{✓}$$

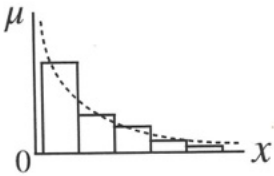
$$\sum_{i=0}^{N-1} \left[\left(\frac{y_i^2}{\mu(x_i | \mathbf{p})^2} - 1 \right) \frac{\partial \mu(x_i | \mathbf{p})}{\partial p_k} \right] = 0$$

Example: $\mu(x_i | \mathbf{p}) = p_0$

$$\sum_{i=0}^{N_b-1} \left(\frac{y_i}{p_0} \right)^2 = \sum 1 = N_b$$

$$\Rightarrow p_0 = \sqrt{\frac{\sum y_i^2}{N_b}}$$

in general, too large $\Rightarrow$ bias!



Poisson vs Gaussian

$$\sum_{i=0}^{N-1} \left[\left(\frac{y_i}{\mu(x_i | \mathbf{p})} - 1 \right) \frac{\partial \mu(x_i | \mathbf{p})}{\partial p_k} \right] = 0$$

$$\sum_{i=0}^{N-1} \left[\left(\frac{y_i^2}{\mu(x_i | \mathbf{p})^2} - 1 \right) \frac{\partial \mu(x_i | \mathbf{p})}{\partial p_k} \right] = 0$$

Example: $\mu(x_i | \mathbf{p}) = p_0/x_i$

$$\sum \frac{y_i x_i}{p_0} \frac{1}{x_i} = \sum \frac{1}{x_i}$$

$$\Rightarrow p_0 = \frac{N_p}{\sum_i (1/x_i)} \Rightarrow p_0 = 100 \text{ as input}$$

E.g. $N_p = 760$, $x_{\min} = \frac{1}{2}$, $x_{\max} = 1000 + \frac{1}{2}$,

$$N_b = 1000, \Delta x = 1 \Rightarrow p(x) = \frac{1/x}{\ln(x_{\max}/x_{\min})}$$

$\Rightarrow$ Poisson: $p_0 \approx 101.5$ (bias due to bhs)

$$N_p \times 10, N_b \times 10 \Rightarrow p_0 \approx 100$$

Example: $\mu(x_i | \mathbf{p}) = p_0/x_i$

$$\sum \left(\frac{y_i^2 x_i^2}{p_0^2} \right) \frac{1}{x_i} = \sum \frac{1}{x_i}$$

$$\Rightarrow p_0 = \sqrt{\frac{\sum y_i^2 x_i}{\sum (1/x_i)}}$$

on average: $p_0 \approx 154$

$$p_0 = 152$$

Different likelihoods and models

- Multiple distributions

If independent: $G \cdot P \cdot P \cdot C \cdot G \dots$

$\Rightarrow -\ln L$ becomes a sum of different terms

- The right space (lin vs log)

— normal dist. vs log-normal

— symmetric vs skewed

— also remove degeneracy; e.g. model $y(x) = a e^{-bx+c}$

↑ bad model: a & c are degenerate
(so remove a or c)

Poisson estimate without bins

$$-\ln \mathcal{L}(\mathbf{p}) = - \sum_{i=0}^{N_b-1} \{y_i \ln[\mu(x_i | \mathbf{p})] - \mu(x_i | \mathbf{p}) - \ln(y_i!)\}$$

← over bins

Choose bins so small that $y_i = 0$ or $y_i = 1$ everywhere

Then: $-\ln \mathcal{L}(\mathbf{p}) = - \sum_{i=0}^{N_p-1} \ln[\mu(x_i | \mathbf{p})] + \int \mu(x | \mathbf{p}) dx + 0$

↖ $\ln(y_i!) = \ln(0!) = \ln(1!)$

Sum over the actual
data values x_i (because $y_i = 1$
only where I have data)

observations,
not bin centers
anymore

integral is often constant, if
model is normalized!
in any case, this integral is
independent of the data

**Next week:
Outliers, model uncertainty
and statistical tests**